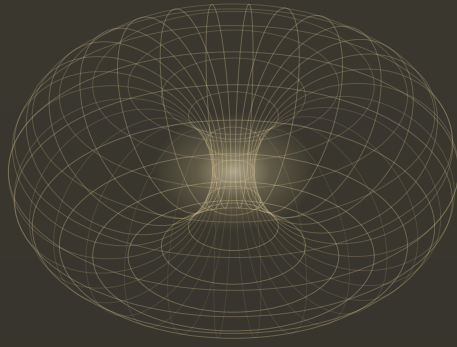




FIELD NOTES

Electroculture, Magnetism and the Toroidal Field

How the CW+ antennas draw the current already moving between Heaven and Earth — and why the shape of that current is a torus.

Electroculture is an agricultural expression of the marriage between Heaven and Earth — an omnipresent, universal circuitry that can be observed as the vivifying force from the cellular level to the macrocosmic. This guide goes a layer deeper: not just that the current exists, but the shape it moves in, and why the CW+ antennas are built the way they are.

I. Energy, not particles

The starting principle beneath everything below: what we're working with is energy in different states, not solid particles bumping around a void. This draws on the field research of Ken Wheeler (Theoria Apophasis), who argues at length that the conventional picture of matter as a collection of discrete particles — electrons included — is a poor model of what's actually happening, and that a single underlying field (he calls it the dielectric field, or simply the aether) is what everything else — magnetism, electricity, gravity — is a temporary expression of. Ice, water and steam are one substance in three states; on this view, magnetism, electricity and gravity are one field in different states.

Why it matters here: if the antenna is working with one continuous field rather than moving discrete "stuff" from one place to another, then the current it draws in from the sky and the telluric current it draws up from the earth aren't two separate resources — they're the same field, expressed two ways.

II. A magnet doesn't emit — it focuses

The CW+ antennas each contain directionally aligned rare earth magnets at their core. Wheeler's central claim about what a magnet actually is applies directly to this component: a magnet doesn't generate or emit a field from nothing. It concentrates and focuses the field already present around it — the same way a magnifying glass doesn't create light, it focuses existing light to a point powerful enough to burn. He places magnets in the same category as lasers: point-source objects that produce a *multiplicative* effect, disproportionate to the raw material involved.

What actually makes something "a magnet," on this model, is not a change in substance but a change in *alignment* — the same atoms, the same mass, before and after magnetising, now moving in unison rather than independently. It's this synchrony, not any added ingredient, that produces the outsized field effect the antenna relies on.

The shape of the field: a torus

Wheeler's other central claim, and the one this whole guide is built around: the true geometry of a magnetic field is toroidal — a donut shape, not the simple two-pole diagram most of us were taught in school. Field-visualising imaging (ferrocell viewing film) shows this same toroidal, spiralling pattern around any magnet, whatever its physical shape. A torus is a single closed loop: field lines exit one pole, arc all the way round through the surrounding space, and re-enter the other pole, then travel back through the material to complete the circuit. Every point on that loop carries both an inbound and an outbound leg at once — nothing on a torus is a dead end.

III. Earth's own field is the same shape

This is where the model stops being speculative and becomes settled, mainstream geophysics: Earth's own magnetic field is a toroidal dipole, generated by the geodynamo in its iron-nickel core. The antenna and its ground connection aren't creating a new field — they're a small-scale version of the same structure Earth already has, tapping into it directly.

A FACT WORTH KNOWING

The Arctic "North Magnetic Pole" is, by strict physics definition, actually a magnetic *south* pole — field lines point into the Earth there. A compass needle's north-seeking tip is drawn toward it precisely because opposite poles attract. The compass isn't pointing at a "north" in the field sense at all: it's a small magnet aligning itself with the much larger toroidal structure of the planet's own field — the same relationship the antenna has with that field. This is also why the Split-Ring Resonator's gap is oriented to the North on installation, and why the ground rod runs north-south: alignment with Earth's own field, not an arbitrary convention.

IV. One current, moving both ways at once

The antenna collects positive cations from the atmosphere. A grounded rod carries the negative anion charge into the earth. Read separately, these can sound like a one-way pipe: charge falls from sky to soil. But a torus doesn't work that way — because it's a single closed loop, *both* ends carry an inbound and an outbound leg simultaneously, not one direction each.

AT THE ANTENNA (THROAT)

Positive cations drawn in and concentrated from the atmosphere — *and*, at the same time, the returning leg of the loop, re-entering here after travelling all the way round via the earth. Both legs, same location.

AT THE GROUND (BLOOM)

Negative anion charge discharging into the rhizome layer — *and*, at the same time, telluric current being drawn back up from the earth via the brass collection rod to feed the other leg of the loop. Both legs, same location.

This is precisely why telluric current collection is already part of the antenna's own design, alongside atmospheric cation collection — not two separate technologies bolted together, but the two legs of one continuous circuit, exactly as a torus requires.

It also matches real, measured atmospheric physics: the fair-weather electric field genuinely has the ionosphere at a positive potential relative to the negatively-charged Earth's surface, driving what's called the global electric circuit — a continuous, two-way current, not a one-way trickle.

Growth by the golden ratio

The two legs of the loop don't simply mirror each other — they scale by the golden ratio ($\phi \approx 1.618$) as they move. The inbound leg concentrates outward from 1 toward ϕ ; the outbound leg contracts back from ϕ toward 1. This isn't decorative: phi has a genuine mathematical property no other ratio has — one divided by phi equals phi minus one ($1/\phi = \phi - 1$) — which means scaling by phi never closes into a repeating, terminating cycle the way a whole-number ratio would. It keeps nesting into itself indefinitely. That's the concrete basis for describing this circuit as continuous rather than something that dissipates or runs down.

One consistent turn

Both legs of the loop, at both ends, share the same rotational handedness — right-turning — so they reinforce rather than cancel one another. This right-handed bias shows up repeatedly in nature: most gastropod shells (snails, whelks, the nautilus) coil dextrally; the double helix of human DNA is right-handed. It's a genuine, well-documented asymmetry in the natural world, not an invented pattern.

V. What each material is actually doing

Real magnetic classification is settled material science, and it maps cleanly onto the CW+ build:

Paramagnetic rock (antenna core fill)	Weakly attracted, receptive — holds no permanent field of its own. The antenna's core and the soil bed around it (see basalt, below) are doing the same kind of work at two different scales.
Rare earth magnets	Genuinely, strongly magnetic — this is the literal point-source object Section II describes: the part of the antenna generating the multiplicative, concentrated field effect.
Copper & brass (body, primary wire, telluric rod)	Diamagnetic — very weakly repelled by a field rather than pinned by one. Its real virtue is exceptional conductivity, letting the concentrated charge keep moving freely along both legs of the loop rather than stalling.
Ground kit (recommended alongside every installation)	Where ferromagnetic (iron-based) materials are used for the earth connection, this is the component coupling most directly into Earth's own iron-core-driven field — completing the circuit's link to the planet at large.

VI. Basalt: widening the field's reach through the whole bed

Basalt's role in electroculture has its own, separate real lineage worth crediting directly: Philip Callahan, an entomologist and agricultural researcher, published measured work on paramagnetism in rocks and soils, developing an actual CGS-unit scale for how paramagnetic different materials are. Basalt — being iron-rich volcanic rock — consistently measures among the higher-paramagnetism rock types in his data.

In the field model above, basalt plays a specific role: not a pole, but a *receptive medium*. Where the ground connection is one concentrated point of contact with Earth's field, basalt broadcast through the surrounding soil extends that same coupling across the whole bed — widening the effective contact area the returning leg of the loop has to work with, rather than confining it to a single point.

Application: roughly 1kg of good-quality basalt rock dust per square metre, broadcast across the growing bed the antenna serves. Electroculture and basalt are meant to be used together, not as alternatives to one another — the antenna concentrates and cycles the field; the basalt gives it a wide surface to couple into.

VII. Sources & further viewing

Chris Winters — The Fertile Current

The CW+ antennas' telluric current collection and atmospheric cation collection — the two legs of the loop this guide describes — are part of the antenna technology built in partnership with Chris Winters of The Fertile Current.

Ken Wheeler — Theoria Apophasis

The toroidal field model, the magnet-as-point-source framework, and the "energy, not particles" foundation in this guide draw directly on Ken Wheeler's independent field research, published across his YouTube channel, Theoria Apophasis. We're grateful to build on his work here.

Philip Callahan

The paramagnetism research behind basalt's role in this model comes from Philip Callahan's published work, *Paramagnetism: Rediscovering Nature's Secret Force of Growth*.

The Living Field — solutionary electroculture. This guide is offered as one layer of explanation among several; see the product pages for the full construction & technology specification of each antenna.